POINT SET TOPOLOGY

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1. METRIC SPACES

1. OPEN AND CLOSED SETS

Definition 1.1 (Metric)**.** Let X be a set. A *metric* on X is a function $d\colon X \times X \to \mathbb{R}$ satisfying the following properties:

1. $d(x, y) \geq 0$ for all $x, y \in X$.
2. $d(x, y) = 0$ if and only if $x = y$ for all $x, y \in X$,
3. $d(x, y) = d(y, x)$ for all $x, y \in X$.
4. $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$. This is often referred to as the *triangle inequality.*

We say X equipped with a metric d is a *metric space*, denoted by the pair (X, d).

Definition 1.2 (Subspace)**.** Let (X, d) be a metric space and let Y be a subset of X. We say that the restriction d_Y of d to $Y \times Y$ is an *induced metric* on Y. We say Y equipped with an induced metric d_Y is a *subspace* of X, denoted by the pair (Y, d_Y).

For the sake of simplicity, going forward we denote every metric space (X, d) as simply X and assume that it is equipped with a metric d.

Definition 1.3 (Open Neighborhood)**.** Let X be a metric space. Let $x \in X$ and $r > 0$. An *open neighborhood* $N_r(x)$ centered at $x \in X$ is the set

$$N_r(x) \coloneqq \{y \in X \colon d(x, y) < r\}.$$

Remark. Notice that every open neighborhood centered at a fixed point $x \in X$ is nested with increasing radius i.e. if $r_2 > r_1 > 0$, then $N_{r_1}(x) \subseteq N_{r_2}(x)$. This is because if $d(x, y) < r_1$ for some $y \in X$ (and thus $y \in N_{r_1}(x)$), then clearly, $d(x, y) < r_2$ as well so $y \in N_{r_2}(x)$.

Definition 1.4 (Interior)**.** Let X be a metric space and Y be a subset of X. We say $y \in Y$ is an *interior point* of Y if there exists $r > 0$ such that $N_r(x) \subseteq Y$.

The set of all interior points of Y is called the *interior* of Y denoted $\operatorname{int}(Y)$, and if every point of Y is an interior point of Y i.e. $\operatorname{int}(Y) = Y$, then we say Y is an *open* subset of X, or that Y is *open* in X.

Remark. Notice that for a point to be an interior point of a subset Y, it must already be in Y, so $\operatorname{int}(Y) \subseteq Y$ is true for any subset Y.

Theorem 1.5. Let X be a metric space. Any open neighborhood $N_r(x)$ of X is an open subset of X.

Proof. Let $y \in N_r(x)$. Then, $d(x, y) < r$. Define $s \coloneqq r - d(x, y)$. Then, for any $z \in N_s(y)$,

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = d(x, y) + r - d(x, y) = r$$

Thus, $z \in N_r(x)$ as well, so $N_s(y) \subseteq N_r(x)$ and y is an interior point of $N_s(x)$. $\square$

Theorem 1.6. Let X be a metric space and Y be a subset of X. Then, the interior of Y is an open subset of X i.e. $\operatorname{int}(Y) = \operatorname{int}(\operatorname{int}(Y))$.

Proof. We already know $\text{int}(\text{int}(Y)) \subseteq \text{int}(Y)$, as we stated previously. Let $y \in \text{int}(Y)$. Then, for some $r > 0$, $N_r(y) \subseteq Y$. By theorem 1.5, we know that $N_r(y)$ is an open subset of Y. Thus, for any $z \in N_r(y)$, there exists $s > 0$ such that $N_s(z) \subseteq N_r(y)$. Thus, every point in $N_r(y)$ is also an interior point of Y, so we conclude that $N_r(y) \subseteq \text{int}(Y)$, and that $\text{int}(Y) \subseteq \text{int}(\text{int}(Y))$. □

Theorem 1.7. Let X be a metric space. The arbitrary union of a family of open subsets of X is an open subset of X.

Proof. Let $\{F_\alpha\}_{\alpha \in A}$ be a family of open subsets of X. Let $x \in \bigcup_{\alpha \in A} F_\alpha$. Then, $x \in F_{\alpha_0}$ for some $\alpha_0 \in A$, and F_{α_0} is open so there exists some $r > 0$ such that $N_r(x) \subseteq F_{\alpha_0}$. Thus, $N_r(x) \subseteq \bigcup_{\alpha \in A} F_\alpha$ so x is an interior point of $\bigcup_{\alpha \in A} F_\alpha$, and we conclude $\bigcup_{\alpha \in A} F_\alpha$ is an open subset of X. □

Theorem 1.8. Let X be a metric space and Y be a subset of X. Y is open in X if and only if Y is the union of open neighborhoods in X.

Proof. ($\Leftarrow$) follows from Theorem 1.7. Now, suppose Y is open in X. Let $y \in Y$. Then, y is an interior point of Y, so there exists $r > 0$ such that $N_r(y) \subseteq Y$. Thus, since for every point $y \in Y$, there exists an open neighborhood centered at y contained in Y, we conclude Y is the union of open neighborhoods in X. □

Theorem 1.9. Let X be a metric space. The finite intersection of open subsets of X is an open subset of X.

Proof. Let $\{G_n\}_{n=1}^N$ be a finite family of open subsets of X. Let $x \in \bigcap_{n=1}^N G_n$. Then, since G_n is open for all $n \in \{1, \ldots, N\}$, x is an interior point of G_n for each n so there exists r_n such that $N_{r_n}(x) \subseteq G_n$ for each n. Let $s \coloneqq \min\{r_n \colon n \in \{1, \ldots, N\}\}$. Then, since open neighborhoods centered at a fixed point are nested with increasing radius i.e. for any $r_n < r_m$, $N_{r_n}(x) \subsetneq N_{r_m}(x)$, then $N_s(x) \subseteq G_n$ for each n, so $N_s(x) \subseteq \bigcap_{n=1}^N G_n$. Thus, x is an interior point of $\bigcap_{n=1}^N G_n$ and we conclude $\bigcap_{n=1}^N G_n$ is an open subset of X. □

Remark. Note the words *arbitrary* and *finite* in Theorem 1.7 and Theorem 1.9, respectively. In general, a finite union of open subsets is not necessarily open, and an infinite intersection of open subsets is not necessarily open.

Theorem 1.10. Let X be metric space and Y be a subspace of X. Then $U \subseteq Y$ is an open subset of Y if and only if $U = V \cap Y$ for an open subset V of X.

Proof. Suppose U is an open subset of Y. Then, by Theorem 1.8, U is the union of open neighborhoods in Y centered at all points $y \in Y$ (with respect to the induced metric d_Y on Y) i.e. $Y = \bigcup_{y \in Y} N_{r_y}(y)$ where $r_y > 0$ is chosen so that $N_{r_y}(y) \subseteq Y$ for each $y \in Y$, respectively. Since each y is in X as well, then by Theorem 1.7, the union of open neighborhoods in X centered at each $y \in Y$, $\bigcup_{y \in Y} N_{r_y}(y)^X$, form an open subset of X. Then, clearly, $U = \bigcup_{y \in Y} N_{r_y}(y) = \bigcup_{y \in Y} (N_{r_y}(y)^X \cap Y) = (\bigcup_{y \in Y} N_{r_y}(y)^X) \cap Y$.

Now, suppose $U = V \cap Y$ for an open subset V of X. Let $y \in U$. Then, y is an interior point of V, so there exists $r > 0$ such that $N_r(y) \subseteq V$. Let $N_r(y)^Y$ denote $N_r(y)$ intersected with Y, so $N_r(y)^Y = \{z \in Y \colon d_Y(y, z) < r\}$ where d_Y is the induced metric on Y. Then, clearly, $N_r(y)^Y \subseteq V \cap Y$, so y is an interior point of U with respect to d_Y. Thus, we conclude that U is open in Y. □

Definition 1.11 (Closure)**.** Let X be a metric space and Y be a subset of X. We say $y \in Y$ is an *adherent point* of Y if for all $r > 0$, $N_r(x) \cap Y \neq \emptyset$.

The set of all adherent points of Y is called the *closure* of Y denoted $\overline{Y}$, and if every adherent point of Y is in Y, then we say Y is a *closed* subset of X, or that Y is *closed* in X.

Remark. Notice that every point of a subset Y is an adherent point of Y, since $y \in N_r(y)$ for all $r > 0$. Thus, $Y \subseteq \overline{Y}$ is true for any subset Y.

We classify adherent points into two categories.

Definition 1.12 (Limit Points, Isolated Points). Let X be a metric space and Y be a subset of X, and let $y \in Y$ be an adherent point of Y.

We say y is a *limit point* of Y if for all $r > 0$, $N_r(x) \cap (Y \setminus \{y\}) \neq \emptyset$.

We say y is a *isolated point* of Y if there exists $r > 0$ such that $N_r(x) \cap (Y \setminus \{y\}) = \emptyset$.

In metric spaces, we are often interested in associating adherent points with limits of convergent sequences.

Definition 1.13 (Convergence). Let X be a metric space, and let $\{x_n\}_{n\geq 1}$ be a sequence in X. We say $\{x_n\}_{n\geq 1}$ converges to some point $x \in X$ if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$d(x_n, x) < \varepsilon$$

Equivalently, we say $\{x_n\}_{n\geq 1}$ converges to some point $x \in X$ if

$$\lim_{n\to\infty} d(x_n, x) = 0$$

Lemma 1.14. In a metric space X, the limit of a convergent sequence is unique.

Proof. Suppose y_1 and y_2 are limits of a convergent sequence $\{x_n\}_{n\geq 1}$ in X. Then,

$$d(y_1, y_2) \leq d(y_1, x_n) + d(x_n, y_2)$$

and as $n \to \infty$, $d(y_1, x_n) + d(x_n, y_2) \to 0$, so $d(y_1, y_2) \to 0$ and $y_1 = y_2$.

□

The following theorem is occasionally treated as a definition of an adherent point in some texts, but we state it as a theorem.

Theorem 1.15. Let X be a metric space and Y be a subset of X. $x \in X$ is an adherent point of Y if and only if a sequence in Y converges to x.

Proof. Suppose $x \in X$ is an adherent point of Y. Then, $N_{\frac{1}{n}}(x) \cap Y \neq \emptyset$ for all $n \geq 1$, so we can construct a sequence $\{x_n\}_{n\geq 1}$ in Y such that $x_n \in N_{\frac{1}{n}}(x) \cap Y$ for each $n \geq 1$. Then, $d(x_n, x) < \frac{1}{n}$ for each n, and as $n \to \infty$, $\frac{1}{n} \to 0$, so $d(x_n, x) \to 0$. Thus, $\{x_n\}_{n\geq 1}$ converges to x.

Now, suppose $\{x_n\}_{n\geq 1}$ is a sequence in Y that converges to some $x \in X$. Then, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $d(x_n, x) < \varepsilon$. Thus, for all $\varepsilon > 0$, $N_\varepsilon(x) \cap Y \neq \emptyset$ so x is an adherent point of Y. □

We deduce that Y is closed if and only if every convergent sequence in Y converges to some point in Y.

Corollary 1.16. Let X be a metric space and Y be a subset of X. $z \in X \setminus Y$ is a adherent point of Y if and only if z is a limit point of Y.

The remaining theorems concern various properties of closed sets, as we did for open sets previously.

Theorem 1.17. Let X be a metric space and Y be a subset of X. Then, the closure of Y is a closed subset of X i.e. $\overline{Y} = \overline{\overline{Y}}$.

Proof. We know $\overline{Y} \subseteq \overline{\overline{Y}}$, as we stated previously. Let $y \in \overline{Y}$. Then, there exists a sequence $\{y_n\}_{n\geq 1}$ in Y that converges to y. We know y_n is an adherent point of Y for all $n \geq 1$, so $y_n \in \overline{Y}$ for each n. Thus, $\{y_n\}_{n\geq 1}$ is a sequence in $\overline{Y}$ that converges to y, so y is an adherent point of $\overline{Y}$ and $y \in \overline{\overline{Y}}$. □

Theorem 1.18. Let X be a metric space and Y be a subset of X. Then, Y is closed if and only if its complement $X \setminus Y$ is open.

Proof. Suppose Y is closed. Then, every convergent sequence $\{y_n\}_{n\geq 1}$ converges to some $y \in Y$. Thus, every point $x \in X \setminus Y$ is not a limit point of Y, so there exists $r > 0$ such that $N_r(x) \cap Y \setminus \{x\} = N_r(x) \cap Y = \emptyset$. Equivalently, $N_r(x) \subset X \setminus Y$. Thus, every point $x \in X \setminus Y$ is an interior point of $X \setminus Y$, so $X \setminus Y$ is open.

Now, suppose Y is open. Then, for all $y \in Y$, there exists $r > 0$ such that $N_r(y) \subseteq Y$. Then, clearly, $N_r(y) \cap (X \setminus Y) = \emptyset$. Thus, no point $y \in Y$ is an adherent point of $X \setminus Y$, which implies that every adherent point of $X \setminus Y$ is in $X \setminus Y$, so $X \setminus Y$ is closed. □

Remark. Note that a set being open does not imply it is not closed, and a set being closed does not imply it is not open! For example, the semiopen interval $[0, 1)$ in $\mathbb{R}$ is neither open or closed in $\mathbb{R}$.

Theorem 1.19. Let X be a metric space. The finite union of closed subsets of X is a closed subset of X.

Proof. Let $\{F_n\}_{n=1}^N$ be a finite family of closed subsets of X. Then, by Theorem 1.18, $X \setminus F_n$ is an open subset of X for each $n \in \{1, \ldots, N\}$. Then, by Theorem 1.9,

$$\bigcap_{n=1}^{N} (X \setminus F_n) = X \setminus (\bigcup_{n=1}^{N} F_n)$$

is an open subset of X, and thus by Theorem 1.17, $\bigcup_{n=1}^{N} F_n$ is a closed subset of X. □

Theorem 1.20. Let X be a metric space. The arbitrary union of closed subsets of X is a closed subset of X.

Proof. Let $\{G_\alpha\}_{\alpha\in A}$ be an arbitrary family of closed subsets of X. Then, by Theorem 1.18, $X \setminus G_\alpha$ is an open subset of X for each $\alpha \in A$. Then, by Theorem 1.9,

$$\bigcup_{\alpha\in A} (X \setminus G_\alpha) = X \setminus (\bigcap_{\alpha\in A} G_\alpha)$$

is an open subset of X, and thus by Theorem 1.17, $\bigcap_{\alpha\in A} G_\alpha$ is a closed subset of X. □

Remark. Again, note the words *finite* and *arbitrary* for Theorem 1.19 and Theorem 1.20, respectively. general, an arbitrary union of closed subsets is not necessarily closed, and a finite intersection of closed subsets is not necessarily closed.

2. COMPLETENESS AND DENSITY

Definition 2.1 (Cauchy sequence)**.** Let X be a metric space and let $\{x_n\}_{n\geq 1}$ be a sequence in X. We say $\{x_n\}_{n\geq 1}$ is a Cauchy sequence if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$,

$$d(x_n, x_m) < \varepsilon$$

Note that while Cauchy sequences in $\mathbb{R}$ always converge as one is likely taught in an introductory real analysis course, in general, this is not necessarily true for metric spaces. In fact, we will we provide more rigorous reasoning as to why Cauchy sequences always converge in $\mathbb{R}$ when treated as a metric space.

Definition 2.2 (Completeness)**.** Let X be a metric space. We say X is *complete* if every Cauchy sequence in X converges in X.

Lemma 2.3. Let X be a metric space. Every convergent sequence $\{x_n\}_{n\geq 1}$ in X is a Cauchy sequence.

Proof. Let $\{x_n\}_{n\geq 1}$ be a sequence in X that converges to some point $x \in X$. Let $\varepsilon > 0$. Then, for $\frac{\varepsilon}{2} > 0$,

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

for all $n, m \in N$ for some $N \in \mathbb{N}$. Thus, $\{x_n\}_{n\geq 1}$ is a Cauchy sequence. □

Lemma 2.4. Let X be a metric space, and let $\{x_n\}_{n\geq 1}$ be a Cauchy sequence in X. If $\{x_n\}_{n\geq 1}$ has a subsequence $\{x_{n_k}\}_{k\geq 1}$ that converges to some point $x \in X$, then $\{x_n\}_{n\geq 1}$ converges to x.

Proof. Let $\{x_n\}_{n\geq 1}$ be a Cauchy sequence in X and suppose it has a subsequence $\{x_{n_k}\}_{k\geq 1}$ that converges to some point $x \in X$. Then, there exists $K \in \mathbb{N}$ such that for all $k \geq K$, $d(x_{n_k}, x) < \frac{\varepsilon}{2}$, and there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $d(x_n, x_m) < \frac{\varepsilon}{2}$ for any $\varepsilon > 0$.

Then, for all $n, n_k \geq \max\{K, N\}$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

so $\{x_n\}_{n\geq 1}$ converges to x. □

Theorem 2.5. $(\mathbb{R}, |\cdot|)$ is a complete metric space.

We give two proofs for this theorem.

Proof. FINISH

□

Theorem 2.6. Let X be a *complete* metric space. If Y is a closed subset of X, then Y is complete.

Proof. Let $\{y_n\}_{n\geq 1}$ be a Cauchy sequence in Y. Then, $\{y_n\}_{n\geq 1}$ is also a Cauchy sequence in X, so it converges to some $x \in X$. However, since Y is closed, any convergence sequence in Y converges in Y, so $\{y_n\}_{n\geq 1}$ converges to some $y \in Y$. Then, since a limit of a convergence sequence is unique, $x = y$, and so $\{y_n\}_{n\geq 1}$ converges in Y. Thus, we conclude that Y is complete. □

Theorem 2.7. Let X be a metric space. If Y is a complete subspace of X, then Y is closed.

Proof. Let $\{y_n\}_{n\geq 1}$ be a sequence in Y that converges. Then, by Lemma 2.3, $\{y_n\}_{n\geq 1}$ is a Cauchy sequence, and since Y is complete, $\{y_n\}_{n\geq 1}$ converges to some $y \in Y$. Thus, Y is closed. □

Definition 2.8 (Dense)**.** Let X be a metric space and let T be a subset of X. We say T is *dense* in X if $\overline{T} = X$. Equivalently, we say T is *dense* in X if T intersects every open subset of X i.e. for any open set U of X, $U \cap T \neq \emptyset$.

Theorem 2.9 (Baire Category Theorem)**.** Let X be a *complete* metric space and let $\{G_n\}_{n\geq 1}$ be a countable family of dense open subsets of X. Then, $\bigcap_{n\geq 1} G_n$ is also dense in X.

Proof. It suffices to show that every open neighborhood in X intersects $\bigcap_{n\geq 1} G_n$ i.e. to show that for any $x \in X$, there exists $\varepsilon > 0$ such that there exists $y \in N_\varepsilon(x)$ such that $y \in \bigcap_{n\geq 1} G_n$. Fix $x \in X$. Then, since G_1 is dense in X, G_1 intersects $N_\varepsilon(x)$, and there exists $y_1 \in G_1$ such that $d(x, y_1) < \varepsilon$ for some $\varepsilon > 0$. Then, since G_1 is open, we can choose $r_1 > 0$ small enough so that that $\overline{N_{r_1}(y_1)} \subseteq G_1 \cap N_\varepsilon(x)$. Following, since G_2 is dense in X, G_2 intersects $N_{r_1}(y_1)$, and there exists $y_2 \in G_2$ such that $d(y_1, y_2) < r_1$. Then, since G_2 is open, we can choose r_2 small enough so that $\overline{N_{r_2}(y_2)} \subseteq G_2 \cap N_{r_1}(y_1)$. Continuing this process for all $n \geq 1$, we have that

$$\overline{N_{r_{n+1}}(y_{n+1})} \subseteq G_{n+1} \cap N_{r_n}(y_n) \tag{1.1}$$

for all $n \geq 1$.

Then, it follows that for any $n \geq 1$,

$$\overline{N_{r_n}(y_n)} \subseteq N_{r_{n-1}}(y_{n-1}) \subseteq \cdots \subseteq N_{r_2}(y_2) \subseteq N_{r_1}(y_1) \subseteq N_\varepsilon(x) \tag{1.2}$$

Then, for any $m > n$, $y_m \in N_{r_n}(y_n)$, and thus, as $r_n \to 0$, $d(y_m, y_n) \to 0$, so we have a Cauchy sequence $\{y_n\}_{n\geq 1}$. Then, since X is Cauchy, $\{y_n\}_{n\geq 1}$ converges to some $y \in X$. Then, since $y_m \in N_{r_n}(y_n)$ for all $m > n$, $y \in \overline{N_{r_n}(y_n)}$. Thus, by (1.2), $y \in N_{r_n}(y_n)$ for all $n \geq 1$. and following by (1.1), $y \in G_n$ for all $n \geq 1$, so $y \in \bigcap_{n\geq 1} G_n$. Thus, we conclude that $\bigcap_{n\geq 1} G_n$ intersects every open neighborhood of X and thus intersects every open subset of X, and thus is dense in X. □

The following two Corollaries are equivalent statements of the Baire Category Theorem.

Definition 2.10 (Nowhere Dense)**.** Let X be a metric space and Y be a subset of X. We say Y is *nowhere dense* if its closure has no interior points i.e. $\text{int}(\overline{Y}) = \emptyset$.

Lemma 2.11. Let X be a metric space. If Y is a nowhere dense subset of X, Then, $X \setminus \overline{Y}$ is a dense open subset of X.

Proof. $X \setminus \overline{Y}$ is open since $\overline{Y}$ is closed.

Let $y \in \overline{Y}$. Then, since $\text{int}(\overline{Y}) = \emptyset$, y is not an interior point of $\overline{Y}$, so for all $r > 0$, $N_r(y) \cap (X \setminus \overline{Y}) \neq \emptyset$, and thus, y is an adherent point of $X \setminus \overline{Y}$. Thus, it follows that $X = \overline{X \setminus \overline{Y}}$, and that $X \setminus \overline{Y}$ is a dense open subset of X. □

Corollary 2.12. Let X be a *complete* metric space and let $\{F_n\}_{n\geq 1}$ be a family of nowhere dense subsets of X. Then, $\bigcup_{n\geq 1} F_n$ has an empty interior.

Proof. Since F_n is nowhere dense for each $n \geq 1$, by Lemma 2.10, $X \setminus \overline{F_n}$ is a dense open subset for each $n \geq 1$. Then, by the Baire Category Theorem,

$$\bigcap_{n\geq 1} (X \setminus \overline{F_n}) = X \setminus (\bigcup_{n\geq 1} \overline{F_n}) = X \setminus \overline{(\bigcup_{n\geq 1} F_n)}$$

is dense and open in X. Thus, $\bigcup_{n\geq 1} F_n$ is nowhere dense and thus has an empty interior. □

Definition 2.13 (First Category, Second Category)**.** Let X be metric space. A subset of X is said to be of the *first category* or *meager* if it is a countable union of nowhere dense sets. A subset of X is said to be of the *second category* if it is not of the first category.

Corollary 2.14. Let X be a *complete* metric space. Then, the complement of a subset of X of the first category is dense in X.

Proof. We demonstrated this in our proof of Corollary 2.11. □

3. PRODUCTS OF METRIC SPACES

Let $(X_1, d_1), (X_2, d_2), \ldots, (X_n, d_n)$ be metric spaces. Then, we can always define a metric on the product set $X = X_1 \times X_2 \times \cdots \times X_n$. While we omit the proofs, the following are examples of metrics on X, where $x, y \in X$ are denoted as $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$:

1. $d(x, y) = [\sum_{k=1}^{n} d_k(x_k, y_k)^2]^{\frac{1}{2}}$
2. $d(x, y) = \max\{d_k(x_k, y_k) \colon k \in \{1, \ldots, n\}\}$
3. $d(x, y) = \sum_{k=1}^{n} d_k(x_k, y_k)$

In metric spaces, we are often interested in metrics that determine the same open sets of a metric space. All three metrics above happen to share the following property, which we will see imply that they determine the same open sets in X:

($*_1$) A sequence $\{x^{(j)}\}_{j\geq 1}$ in X converges to some $x = (x_1, x_2, \ldots, x_n) \in X$ if and only if the sequence in each component $\{x_k^{(j)}\}_{j\geq 1}$ converges to x_k for each $k \in \{1, \ldots, n\}$.

Theorem 3.1. Let $X = X_1 \times X_2 \times \cdots \times X_n$ and let d be a metric on X that satisfies ($*_1$). Then, the open sets in (X, d) are the union of product sets of the form $U_1 \times U_2 \times \cdots \times U_n$, where U_k is an open subset of X_k for each $k \in \{1, \ldots, n\}$.

Proof. Let $F_1, F_2, \ldots, F_n$ be closed subsets of $X_1, X_2, \ldots, X_n$, respectively. We claim $F = F_1 \times F_2 \times \cdots \times F_n$ is closed. Let $\{x^{(j)}\}_{j\geq 1}$ be a convergent sequence in F. Then, by ($*_1$), the sequence in each component $\{x_k^{(j)}\}_{j\geq 1}$ converges to $x_k \in F_k$ since F_k is closed for each $k \in \{1, \ldots, n\}$. Thus, $\{x^{(j)}\}_{j\geq 1}$ converges to $x = (x_1, x_2, \ldots, x_n) \in F$, so F is closed.

Now, let $U_1, U_2, \ldots, U_n$ be open subsets of $X_1, X_2, \ldots, X_n$, respectively. Then, for each $k \in \{1, \ldots, n\}$, $X_k \setminus U_k$ is a closed subset of X_k. Then, for each $k \in \{1, \ldots, n\}$,

$$X_1 \times X_2 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n$$

is a closed subset of X. Then, its complement

$$V^{(k)} := X_1 \times X_2 \times \cdots \times U_k \times \cdots \times X_n$$

is an open subset of X. Then,

$$\bigcap_{k=1}^{n} V^{(k)} = U_1 \times U_2 \times \cdots \times U_k$$

is an open subset of X.

Now, to show that if U is an open subset of X, then $U = U_1 \times U_2 \times \cdots \times U_n$ for open subsets $U_1, U_2, \ldots, U_n$ of $X_1, X_2, \ldots, X_n$, respectively, it suffices to show that if $x \in U$, $x \in U_1, U_2, \ldots, U_n$.

Arguing by contradiction, suppose $x \notin U_1, U_2, \ldots, U_n$ for any such open subsets. Then, the product of open neighborhoods centered at x_k for each $k \in \{1, \ldots, n\}$ with radius $\frac{1}{m}$ for each $m \geq 1$, $N_{\frac{1}{m}}(x_1) \times N_{\frac{1}{m}}(x_2) \times \cdots \times N_{\frac{1}{m}}(x_2)$, is not contained in U. Then, for some $\{x^{(j)}\}_{j\geq 1} \in X \setminus U$, there exists

$$d_k(x_k^{(j)}, x_k) < \frac{1}{m}$$

for each $k \in \{1, \ldots, n\}$. Thus, $\{x^{(j)}\}_{j\geq 1}$ converges to x, but this is a contradiction since U is open so $X \setminus U$ is closed, but $\{x^{(j)}\}_{j\geq 1}$ converges in U. Thus, we conclude that U is the product of open subsets of X_k for each $k \in \{1, \ldots, n\}$. □

In addition, we see that all three metrics from the earlier example satisfy another property:

$$(*_2)\ d(x_k, y_k) \leq d(x, y) \text{ for all } x, y \in X \text{ and } k \in \{1, \ldots, n\}.$$

Theorem 3.2. Let $(X_1, d_1), (X_2, d_2), \ldots, (X_n, d_n)$ be complete metric spaces. Let d be a metric on $X = X_1 \times X_2 \times \cdots \times X_n$ that satisfies $(*_1)$ and $(*_2)$. Then, (X, d) is complete.

Proof. Let $\{x^{(j)}\}_{j\geq 1}$ be a Cauchy sequence in X. Then, $d_k(x_k^{(i)}, x_k^{(j)}) \leq d(x^{(i)}, x^{(j)})$, so for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i, j \geq N$, we have $d_k(x_k^{(i)}, x_k^{(j)}) \leq d(x^{(i)}, x^{(j)}) < \varepsilon$, so $\{x_k^{(j)}\}_{j\geq 1}$ is Cauchy for each $k \in \{1, \ldots, n\}$. Then, since X_k is complete, $\{x_k^{(j)}\}_{j\geq 1}$ converges for each $k \in \{1, \ldots, n\}$. Thus, $\{x^{(j)}\}_{j\geq 1}$ converges. □

Corollary 3.3. $(\mathbb{R}^n, ||\cdot||)$ is complete.

4. COMPACTNESS

Definition 4.1 (Open Cover)**.** Let X be a metric space and let S be a subset of X. We say that a family of open subsets $\{G_\alpha\}_{\alpha\in A}$ is an *open cover* of S if $S = \bigcup_{\alpha\in A} G_\alpha$. In general, we can omit the words "open" in the above sentence to say that $\{G_\alpha\}_{\alpha\in A}$ is simply a *cover* of S or that it *covers* S.

Definition 4.2 (Covering Compactness)**.** Let X be a metric space. We say that X is *covering compact* or just *compact* if every open cover of X has a finite subcover i.e. there exists a finite subset of every open cover that covers X.

Definition 4.3 (Totally Bounded)**.** Let X be a metric space. We say that X is *totally bounded* if for all $\varepsilon > 0$ finitely many open neighborhoods with radius $\varepsilon > 0$ cover X.

Definition 4.4 (Sequential Compactness)**.** Let X be a metric space. We say that X is *sequential compact* if every sequence in X has a convergent subsequence.

Theorem 4.5. Let X be a metric space. The following statements are equivalent:

1. X is covering compact.
2. X is sequentially compact.
3. X is totally bounded and complete.

This theorem encapsulates the overall conclusion we aim to attain in this section. In order to prove it, we separate it into Lemmas of much smaller scale.

Lemma 4.6. Let X be a metric space. If X is covering compact, it is sequentially compact.

Proof. Suppose X is covering compact. Let $\{y_n\}_{n\geq 1}$ be a sequence in X.

First, we claim that there exists $x \in X$ such that the neighborhood $N_\varepsilon(x)$ contains infinitely many terms of $\{y_n\}_{n\geq 1}$. Arguing by contradiction, suppose that for all $x \in X$, there exists $\varepsilon_x > 0$ so that $N_{\varepsilon_x}(x)$ contains only finitely many terms of $\{y_n\}_{n\geq 1}$. We see that $\{N_{\varepsilon_x}(x)\}_{x\in X}$ is an open cover of X, and since X is covering compact, there exists a finite subcover $\{N_{\varepsilon_{x_n}}(x_n)\}_{n=1}^{N}$. Then, since each neighborhood contains only finitely many terms of $\{y_n\}_{n\geq 1}$, then the finite subcover contains only finite many terms of $\{y_n\}_{n\geq 1}$, a contradiction! Thus, we conclude that there exists $x \in X$ such that the neighborhood $N_\varepsilon(x)$ contains infinitely many terms of $\{y_n\}_{n\geq 1}$.

Fix $x_0 \in X$ such that there exists $\varepsilon_0 > 0$ such that $N_{\varepsilon_0}(x_0)$ contains infinitely many terms of $\{y_n\}_{n\geq 1}$. Then, we can find a term $n_1 \geq 1$ such that $y_{n_1} \in N_{\varepsilon_0}(x_0)$, then $n_2 \geq n_1$ such that $y_{n_2} \in N_{\frac{\varepsilon_0}{2}}(x_0)$, and continuing for all $k \geq 1$, $n_k \geq n_{k-1} \geq \cdots \geq n_2 \geq n_1$ such that $y_{n_k} \in N_{\frac{\varepsilon_0}{k}}(x_0)$.

Then, we have a subsequence $\{y_{n_k}\}_{k\geq 1}$ such that for each $K \in \mathbb{N}$, for all $k \geq K$,

$$d(y_{n_k}, x) < \frac{\varepsilon_0}{K}$$

and clearly as $K \to \infty$, $\frac{\varepsilon_0}{K} \to 0$, so $d(y_{n_k}, x) \to 0$ so $\{y_{n_k}\}_{k\geq 1}$ converges to x. Thus, we conclude that every sequence in X has a convergent subsequence. □

The above Lemma proves (4.5.1) ⇒ (4.5.2).

Lemma 4.7. Let X be a metric space. If X is sequentially compact, it is totally bounded and complete.

Proof. Suppose X is sequentially compact. Let $\{x_n\}_{n\geq 1}$ be a Cauchy sequence in X. Then, any subsequence of $\{x_n\}_{n\geq 1}$ is also Cauchy and also converges since X is sequentially compact. Then, since there exists a Cauchy subsequence that converges, $\{x_n\}_{n\geq 1}$ also converges, and thus, we conclude that X is complete.

Arguing by contradiction, suppose there exists $\varepsilon > 0$ such that there exists no finite subcover of $\{N_\varepsilon(x)\}_{x\in X}$. Then, we can construct a sequence $\{y_n\}_{n\geq 1}$ in X by the following: Let $y_1 \in X$ such that $N_\varepsilon(y_1) \neq X$. Then, let $y_2 \in X \setminus N_\varepsilon(y_1)$. Continuing let $y_k \in X \setminus (\bigcup_{n=1}^{k-1} N_\varepsilon(y_n))$ for all $k \geq 1$. Then, we have $\{y_n\}_{n\geq 1}$ is a sequence such that $d(y_n, y_m) \geq \varepsilon$ for all $n \neq m$, so $\{y_n\}_{n\geq 1}$ has no convergent subsequence, a contradiction! Thus, we conclude that there exists a finite subcover of $\{N_\varepsilon(x)\}_{x\in X}$. □

The above Lemma proves (4.5.2) ⇒ (4.5.3).

Lemma 4.8. Let X be a totally bounded metric space. Then, every sequence of X has a *Cauchy* subsequence.

Proof. Diagonalisation argument. FINISH □

Notice that if X is complete, then the above statement is equivalent to the statement that every sequence of X has a convergent subsequence, since every Cauchy subsequence would converge. Thus, **the above Lemma proves (4.5.3) ⇒ (4.5.2).**

Definition 4.9 (Separable)**.** Let X be a metric space. We say X is *separable* if there exists a countable dense subset of X.

Lemma 4.10. A totally bounded metric space is separable.

Proof. Let X be a totally bounded metric space. Then, for each $n \geq 1$, there exists a finite subcover of the open cover $\{N_{\frac{1}{n}}(x)\}_{x \in X}$, and we denote the centers of the neighborhoods in the subcover as $x_{n,1}, x_{n,2}, \dots, x_{n,d}$. We repeat this process for all $n \geq 1$, and since a countable union of finite sets is countable, then we have a sequence $\{x_{n,k}\}$ of X. We constructed this sequence so that for any $x \in X$, there exists $x_{n,k}$ such that $d(x_{n,k}, x) < \frac{1}{n}$ for all $n \geq 1$. Thus, we have that $\overline{\{x_{n,k}\}} = X$ and we conclude that X is separable. □

Definition 4.11 (Base)**.** Let X be a metric space. We say a set of open subsets $\mathfrak{B}$ is a *base* of X if for any open subset V of X such that $x \in V$, there exists $A \in \mathfrak{B}$ such that $x \in A \subseteq V$.

Definition 4.12 (Second Countable)**.** Let X be a metric space. We say X is *second countable* if there exists a base of X that is at most countable.

Lemma 4.13. A metric space is second countable if and only if it is separable.

Proof. Suppose X is a second countable metric space. Then, there exists a base $\{G_n\}_{n\geq 1}$ of X that is at most countable. Construct a sequence $\{x_n\}_{n\geq 1}$ where $x_n \in G_n$ for all $n \geq 1$. Then, see that for any nonempty open subset U of X, there exists $x_n \in U$, and we deduce that $\{x_n\}_{n\geq 1}$ intersects any open subset of X so it is dense in X. Thus, since X has a countable dense subset, is separable.

Now, suppose X is separable, so there exists a countable dense subset $\{x_n : n \geq 1\}$ of X. Now, consider the family of open subsets $\{N_{\frac{1}{k}}(x_n) : n \geq 1, k \geq 1\}$. Let U be an open subset of X and let $x \in U$ for some $x \in X$. Then, since x is an interior point of U, there exists some $n \geq 1$ such that $N_{\frac{1}{2n}}(x) \subseteq U$. □

Lemma 4.14 (Lindelöf's Theorem)**.** Let X be a metric space. If X is second countable, then every open cover of X has a *countable* subcover.

Proof. Suppose X is second countable. Then, let $\mathfrak{B}$ be a countable base of X and let $\{G_\alpha\}_{\alpha\in A}$ be an open cover of X. Then, for any $\alpha \in A$, if $x \in G_\alpha$, there exists $S \in \mathfrak{B}$ such that $x \in S \subseteq G_\alpha$, and we denote $\mathfrak{C}$ as the set of all such S. We deduce that $\mathfrak{C}$ is an open cover of X as well. Then, for any $S \in \mathfrak{C}$, let $\alpha_S \in A$ denote an index for which $S \subseteq G_{\alpha_S}$, and let $A' \subseteq A$ denote the set of all such indices. Then, since $\mathfrak{C}$ covers X, it is clear that $\{G_\alpha\}_{\alpha\in A'}$ also covers X, and we conclude that every open cover of X has a countable subcover. □

To summarize, from Lemmas 4.10, 4.13, 4.14, we conclude that:

Totally bounded ⇒ Separable ⇒ Second countable ⇒ Every open cover has countable subcover.

Lemma 4.15. Let X be a metric space. If X is totally bounded and complete, then X has is covering compact.

Proof. Suppose X is totally bounded and complete, so we deduce that every open cover of X has a finite subcover, and also that X is also sequentially compact. Let $\{G_\alpha\}_{\alpha\in A}$ be an open cover of X.

Arguing by contradiction, suppose that $\{G_\alpha\}_{\alpha\in A}$ does not have a finite subcover i.e. $X \neq \bigcup_{n=1}^k G_{\alpha_n}$ for any $k \in \mathbb{N}$. We construct a sequence $\{x_k\}_{k\geq 1}$ where $x_k \in X \setminus (\bigcup_{n=1}^k G_{\alpha_n})$ for every $k \geq 1$. By assumption, $\{x_k\}_{k\geq 1}$ has a convergent subsequence that converges to some $x \in X$. Note that for all $j \geq k$ for any $k \geq 1$, $x_j \in X \setminus (\bigcup_{n=1}^k G_{\alpha_n})$, and since $X \setminus (\bigcup_{n=1}^k G_{\alpha_n})$ is closed, $x \in X \setminus (\bigcup_{n=1}^k G_{\alpha_n})$. Then, we deduce that $\bigcup_{n=1}^k G_{\alpha_n}$ does not cover X for any $k \geq 1$, so $\{G_\alpha\}_{\alpha\in A}$ does not have a countable subcover, a contradiction! Thus, we conclude that every open cover of X has a finite subcover, and thus that X is covering compact. □

The above Lemma proves (4.5.3) ⇒ (4.5.1) *and* (4.5.2) ⇒ (4.5.1).

Corollary 4.16. Let X be a metric space. If X is covering compact, then X is closed and bounded.

Proof. Since X is sequentially compact as well, then every sequence in X has a subsequence that converges in X. Thus, every convergent sequence also converges in X, so X is closed.

X is also totally bounded as well. Let $\varepsilon > 0$, and let $\{N_\varepsilon(x_n)\}_{n=1}^N$ be a finite subcover of $\{N_\varepsilon(x)\}_{x\in X}$. Then, there exists $M := \max\{d(x_m, x_n) : m, n \in \{1, \ldots, N\}\}$. Then, for m, n such that $d(x_m, x_n) = M$, let $y_m \in N_\varepsilon(x_m)$, $y_n \in N_\varepsilon(x_n)$. Notice that

$$d(y_m, y_n) \leq d(y_m, x_m) + d(x_m, x_n) + d(x_n, y_n) < \varepsilon + M + \varepsilon = 2\varepsilon + M$$

Thus, we conclude that X is bounded by $2\varepsilon + M$. □

Remark. In general, the converse of the above Corollary is not necessarily true for metric spaces i.e. a metric space being closed and bounded *does not* imply that it is covering compact (or sequentially compact). However, the converse *is* true for $\mathbb{R}^n$, as detailed below.

We establish an order relation for $\mathbb{R}^n$. For any $a, b \in \mathbb{R}^n$ denoted $a = (a_1, a_2, \ldots, a_n)$ and $b = (b_1, b_2, \ldots, b_n)$, we say $a < b$ if $a_k < b_k$ for all $k \in \{1, \ldots, n\}$. We omit the proof for this order relation.

Definition 4.17 (n-cell)**.** Let $a, b \in \mathbb{R}^n$ such that $a < b$. We say that the set of all $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ such that $a_k \leq x_k \leq b_k$ for all $k \in \{1, \ldots, n\}$ is an *n-cell.*

Note that a 1-cell (when $n = 1$) is simply a closed interval $[a, b]$ in $\mathbb{R}$.

Lemma 4.18. Let $\{I_j\}_{j\geq 1}$ be a family of 1-cells (or closed intervals) such that $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots \supseteq I_{j+1} \supseteq I_j$ for all $j \geq 1$. Then, $\bigcap_{j\geq 1} I_j \neq \emptyset$.

Proof. Since I_j is a closed interval, we have that for some $a_j, b_j \in \mathbb{R}$ for all $j \geq 1$, $I_1 = [a_1, b_1], I_2 = [a_2, b_2], \ldots, I_j = [a_j, b_j]$, and so on. $\{a_j\}_{j\geq 1}$ is monotonically decreasing and $\{b_j\}_{j\geq 1}$ is monotonically increasing, and since both sets are bounded, $\{a_j\}_{j\geq 1}$ converges to some $A \in \mathbb{R}$ and $\{b_j\}_{j\geq 1}$ converges to some $B \in \mathbb{R}$. Since $a_j < b_j$ for all $j \geq 1$, $A \leq b_j$, $a_j \geq B$ for all $j \geq 1$, so $A \leq B$. Thus, we conclude that $[A, B] \subseteq [a_j, b_j] = I_j$ for all $j \geq 1$ and that $\bigcap_{j\geq 1} I_j \neq \emptyset$. □

Lemma 4.19. Let $\{I^{(j)}\}_{j\geq 1}$ be a family of n-cells such that $I^{(1)} \supseteq I^{(2)} \supseteq \cdots \supseteq I^{(j+1)} \supseteq I^{(j)}$ for all $j \geq 1$. Then, $\bigcap_{j\geq 1} I^{(j)} \neq \emptyset$.

Proof. Note that each component $I_k^{(j)}$ for each $j \geq 1$ and $k \in \{1, \ldots, n\}$ is a 1-cell, and $I_k^{(1)} \supseteq I_k^{(2)} \supseteq \cdots \supseteq I_k^{(j+1)} \supseteq I_k^{(j)}$ for all $j \geq 1$. By Lemma 4.18, $\bigcap_{j\geq 1} I_k^{(j)} \neq \emptyset$ for each $k \in \{1, \ldots, n\}$ and thus, $\bigcap_{j\geq 1} I^{(j)} \neq \emptyset$. □

Theorem 4.20 (Heine-Borel Theorem)**.** Let E be a subset of $\mathbb{R}^n$. The following statements are equivalent:

1. E is compact.
2. E is sequentially compact.
3. E is closed and bounded.

It suffices to prove that (4.17.3) ⇒ (4.17.1).

Lemma 4.21. Let E be a subset of $\mathbb{R}^n$. If E is closed and bounded, it is compact.

Proof. Suppose $E \subseteq \mathbb{R}^n$ is closed and bounded. Then, for any $x, y \in E$,

$$d_2(x, y) = [\sum_{k=1}^n (x_k - y_k)^2]^{\frac{1}{2}} \leq \delta$$

for some $\delta \in \mathbb{R}$. Notice that E is an n-cell.

Arguing by contradiction, suppose there exists an open cover of $\{G_\alpha\}_{\alpha\in A}$ of E with no finite subcover. Since E is an n-cell, for each $k \in \{1, \ldots, n\}$, we have a component $E_k = [a_k, b_k]$ for some $a_k, b_k \in \mathbb{R}$. Then, let $c_k^{(1)} = \frac{b_k - a_k}{2}$, and divide $[a_k, b_k]$ into $[a_k, c_k^{(1)}] \cup [c_k^{(1)}, b_k]$ for each

$k \in \{1, \ldots, n\}$. Then, since $\{G_\alpha\}_{\alpha \in A}$ has no finite subcover, there exists $k \in \{1, \ldots, n\}$ such that $[a_k, c_k^{(1)}]$ or $[c_k^{(1)}, b_k]$ is not finitely covered, Without loss of generality, suppose that $[a_k, c_k^{(1)}]$ is not finitely covered, and let $I^{(1)}$ denote the n-cell with the component $[a_k, c_k^{(1)}]$.

Continuing the process, then divide $[a_k, c_k^{(1)}]$ at $c_k^{(2)} = \frac{c_k^{(1)} - a_k}{2}$ into $[a_k, c_k^{(2)}] \cup [c_k^{(2)}, c_k^{(1)}]$ for each $k \geq 1$. Without loss of generality, suppose that $[a_k, c_k^{(2)}]$ is not finitely covered for some $k \in \{1, \ldots, n\}$, and let $I^{(2)}$ denote the n-cell with the component $[a_k, c_k^{(2)}]$. Continuing this process, we have that for each $k \in \{1, \ldots, n\}$, $\operatorname{diam}(I_k^{(i)}) = \frac{b-a}{2^i}$ for each $i \geq 1$, and $I^{(1)} \supseteq I^{(2)} \supseteq \cdots \supseteq I^{(i)}$ for all $i \geq 1$. Then, by Lemma 4.19, $\bigcap_{i \geq 1} I^{(i)} \neq \emptyset$. Then, there exists some $x \in \bigcap_{i \geq 1} I^{(i)}$, and $x \in G_{\alpha_0}$ for some $\alpha_0 \in A$. Since G_{α_0} is open, there exists some $r > 0$ such that $N_r(x) \subseteq G_{\alpha_0}$, and as $i \to \infty$, $\operatorname{diam}(I_k^{(i)}) \to 0$, so there exists i large enough so that $I^{(i)} \subseteq N_r(x) \subseteq G_{\alpha_0}$. This is a contradiction, since we constructed $I^{(i)}$ for each $i \geq 1$ to not be finitely covered! Thus, we conclude that every open cover of E has a finite subcover and that E is covering compact. □

5. CONTINUITY

Definition 5.1 (Continuity). Let (X_1, d_1) and (X_2, d_2) be metric spaces. We say a function $f\colon X_1 \to X_2$ is *continuous* at some point $x \in X_1$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $y \in X$ such that $d_1(x, y) < \delta$, $d_2(f(x), f(y)) < \varepsilon$.

If f is continuous at all points in X, we say f is continuous on X.

Definition 5.2 (Uniform Continuity). Let (X_1, d_1) and (X_2, d_2) be metric spaces. We say a function $f\colon X_1 \to X_2$ is *uniformly continuous* on X if for all $\varepsilon > 0$, there exists $\delta > 0$ such that *for all* $x, y \in X$, if $d_1(x, y) < \delta$, then $d_2(f(x), f(y)) < \varepsilon$.

The following is treated as a definition in some texts, but we prove it as a theorem.

Theorem 5.3. Let (X_1, d_1) and (X_2, d_2) be metric spaces. $f\colon X_1 \to X_2$ is continuous at some point $x \in X_1$ if and only if for all sequences $\{x_n\}_{n \geq 1}$ that converge to x, $\{f(x_n)\}_{n \geq 1}$ converges to $f(x)$.

Proof. Suppose f is continuous. Let $\{x_n\}_{n \geq 1}$ be a sequence in X that converges to some $x \in X$. Let $\varepsilon > 0$. Since f is continuous, there exists some $\delta > 0$ such that if $d_1(x_n, x) < \delta$, then, $d_2(f(x_n), f(x)) < \varepsilon$. Since $\{x_n\}_{n \geq 1}$ converges to x, there exists some $N \in \mathbb{N}$ such that for all $n \geq N$, $d_1(x_n, x) < \delta$, so similarly, for all $n \geq N$, $d_2(f(x_n), f(x)) < \varepsilon$, so $\{f(x_n)\}_{n \geq 1}$ converges to $f(x)$.

Now, suppose that for all sequences $\{x_n\}_{n \geq 1}$ that converge to x, $\{f(x_n)\}_{n \geq 1}$ converges to $f(x)$. Arguing by contradiction, suppose f is not continuous. Then, there exists $\varepsilon > 0$ such that for all $\delta > 0$, if $d_1(x_n, x) < \delta$ then $d_2(f(x_n), f(x)) \geq \varepsilon$. Since $\{x_n\}_{n \geq 1}$ converges, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $d_1(x_n, x) < \delta$. However, for any $N \in \mathbb{N}$, there exists $n \geq N$ such that $d_2(f(x_n), f(x)) \geq \varepsilon$, so $\{f(x_n)\}_{n \geq 1}$ does not converge, a contradiction! □

FINISH

2. TOPOLOGICAL SPACES

1. TOPOLOGICAL SPACES

Definition 1.1 (Topology)**.** Let X be a set. We say $\mathcal{T}$, a family of subsets of X is a *topology* for X is $\mathcal{T}$ satisfies the following:

1. $X, \emptyset \in \mathcal{T}$.
2. Any union of subsets of $\mathcal{T}$ is in $\mathcal{T}$.
3. Any finite intersection of subsets of $\mathcal{T}$ is in $\mathcal{T}$.

We say that the pair $(X, \mathcal{T})$ is a *topological space.* Differing from the respective definitions for a metric space, we say a subset of X is *open* if it is $\mathcal{T}$, and we say a subset of X is *closed* if its complement is open.

For the sake of simplicity, going forward we denote every topological space $(X, \mathcal{T})$ as simply X and assume that it has a topology $\mathcal{T}$.

The following are some examples of topologies:

1. The *discrete topology* consists of all subsets of X.
2. The *indiscrete topology* consists of only X and $\emptyset$.
3. If X is a metric space, the *metric topology* consists of all subsets of X that are open with respect to a metric on X.

Definition 1.2 (Metrizable)**.** Let X be a topological space. We say X is *metrizable* if the topology for X is the metric topology associated with some metric on X i.e. the subsets of X in the topology are open with respect to some metric on X.

Definition 1.3 (Neighborhood)**.** Let X be a topological space. We say a subset S of X is a *neighborhood* of $x \in X$ if there exists some open set U such that $x \in U \subseteq S$.

Definition 1.4 (Interior)**.** Let X be a topological space and let S be a subset of X. We say $x \in S$ is an *interior point* of X if S is a neighborhood of X. We say that the set of all interior points of S is the *interior* of S, denoted by $\operatorname{int}(S)$.

Theorem 1.5. Let X be a topological space. A subset S of X is open if and only if $S = \operatorname{int}(S)$ i.e. S is a neighborhood of all of its points.

Proof. Suppose S is open. Then, for any $x \in S$, S is by definition a neighborhood of x so x is an interior point of S. Thus, $\operatorname{int}(S) = S$.

Now, suppose S is a neighborhood of all of its points. Then, for any $x \in S$, there exists some open set U_x such that $x \in U_x \subseteq S$. Then, clearly, $\bigcup_{x \in S} U_x = S$ and $\bigcup_{x \in S} U_x$ is open, so S is open. $\square$

Theorem 1.6. Let X be a topological space and let S be a subset of X. $\operatorname{int}(S)$ is open i.e. $\operatorname{int}(S) = \operatorname{int}(\operatorname{int}(S))$.

Proof. Let $x \in \text{int}(S)$. Then, there exists some open set U_x such that $x \in U \subseteq S$. Notice that for any $y \in U_x$, y is an interior point of S as well. Thus, $x \in U_x \subseteq \text{int}(S)$, so x is an interior point of $\text{int}(S)$. Thus, by Theorem 1.5, we conclude that $\text{int}(S)$ is open. $\square$

Definition 1.7 (Closure)**.** Let X be a topological space and let S be a subset of X. We say $x \in S$ is an *adherent point* of S if every neighborhood of X intersects S. We say that the set of all adherent points of S is the *closure* of S, denoted by $\overline{S}$.

Theorem 1.8. Let X be a topological space and S be a subset of X. S is closed if and only if $S = \overline{S}$.

Proof. Suppose S is closed. Then, $X \setminus S$ is open. Then, for every $x \in X \setminus S$, there exists an open set U such that $x \in U \subseteq X \setminus S$. Thus, x is not an adherent point of S. Then, since it is clear that every point in S is an adherent point of S, we conclude that $S = \overline{S}$.

Now, suppose $S = \overline{S}$. Then, every point $x \in X \setminus S$ is not an adherent point of S. Thus, there exists some open set U of x that does not intersect S i.e. $S \in X \setminus S$. Thus, x is an interior point of $X \setminus S$ and we conclude that $X \setminus S$ is open, so then S is closed. $\square$

Theorem 1.9. Let X be a topological space and let S be a subset of X. $\overline{S}$ is closed i.e. $\overline{S} = \overline{\overline{S}}$.

Proof. Let $x \in X \setminus \overline{S}$. Then, x is not an adherent point of S i.e. there exists some neighborhood U of x such that $x \in U \subseteq X \setminus \overline{S}$. Notice that for any $y \in U$, U is a neighborhood of y, so y is also an interior point of U. Thus, we conclude that every point in $X \setminus \overline{S}$ is an interior point of $X \setminus \overline{S}$ so $X \setminus \overline{S}$ is open, and thus that $\overline{S}$ is closed. $\square$

Next, we introduce the notion of convergent sequences in topological spaces.

Definition 1.10 (Convergence)**.** Let X be a topological space and let $\{x_n\}_{n\geq 1}$ be a sequence in X. We say $\{x_n\}_{n\geq 1}$ *converges* to some point $x \in X$ if for every neighborhood U of x, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$.

Theorem 1.11. Let X be a topological space and let S be a subset of X. If $\{x_n\}_{n\geq 1}$ is a sequence in S and it converges to some point $x \in X$, then $x \in \overline{S}$.

Proof. Let U be an open neighborhood of x. Then, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$. Thus, we conclude that every open neighborhood of x intersects S, so x is an adherent point of S and thus $x \in \overline{S}$. $\square$

Remark. Note that unlike in metric spaces, in general, the converse of the above statement is *not* true in topological spaces. However, it is true for any metrizable topological space.

Definition 1.12 (Boundary)**.** Let X be a topological space and let S be a subset of X. We say $x \in X$ is a *boundary point* of S if x is an adherent point of both S and $X \setminus S$. The *boundary* of S is the set of all boundary points of S, denoted by ∂S.

Notice that

$$\partial S = \overline{S} \cap \overline{X \setminus S}$$

and since $\overline{S}$ and $\overline{X \setminus S}$ are closed, then ∂S is closed.

Theorem 1.13. Let X be a topological space and let S be a subset of X. $\overline{S}$ is the disjoint union of $\text{int}(S)$ and ∂S.

Proof. Let $x \in \overline{S}$. Then, every open neighborhood of x intersects S. We have two disjoint cases:

1. If there exists an open neighborhood of x that is contained in S i.e. does not intersect $X \setminus S$, then x is an interior point of S, and thus $x \in \text{int}(S)$.

2. If every open neighborhood of x intersects $X \setminus S$, then x is an adherent point of $X \setminus S$ as well, so $x \in \partial S$.

Thus, we conclude that $\overline{S}$ is the disjoint union of $\text{int}(S)$ and $\partial(S)$. $\square$

Definition 1.14 (Subspace)**.** Let $(X, \mathcal{T})$ be a metric space and let S be a subset of X. We say that the family $\mathcal{S}$ of subsets of S,

$$\mathcal{S} = \{U \cap S : U \in \mathcal{T}\}$$

is the *relative topology* for S inherited from X. The pair $(S, \mathcal{S})$ is called a *subspace* of $(X, \mathcal{T})$ and we say every subset $V \in \mathcal{S}$ a *relatively open* subset of S.

Remark. Note that every open subset of S is not necessarily open in X.

2. CONTINUOUS FUNCTIONS

As in the previous section, the definition of continuity varies slightly between metric spaces and topological spaces. In particular, the definitions of being continuous on a whole space or a subset of a whole space varies slightly from the definition of being continous at a singular point.

Definition 2.15 (Continuity)**.** Let X and Y be topological spaces. A function $f: X \to Y$ is *continuous* if for any open subset U of Y, $f^{-1}(U)$ is an open subset of X. We also say that f is a *map* from X to Y.

Definition 2.16 (Continuity at a Point)**.** Let X and Y be topological spaces. A function $f: X \to Y$ is *continuous* at some point $x \in X$ if for every open subset V of Y such that $f(x) \in V$, there exists an open subset U of X such that $x \in U$ and $f(U) \subseteq V$.

Theorem 2.17. Let X and Y be topological spaces. A function $f: X \to Y$ is continuous if and only if f is continuous at every point of X.

Proof. Suppose that f is continuous. Let $x \in X$ and let V be an open subset of Y such that $f(x) \in V$. Since f is continuous, $U = f^{-1}(V)$ is an open subset of X containing x and $f(U) \subseteq V$, so f is continuous at x.

Now, suppose that f is continuous at every point of X, and let V be an open subset of Y. Let $x \in f^{-1}(V)$. Since f is continuous at x, there exists an open subset U_x of X such that $x \in U_x$ and $f(U_x) \subseteq V$. We deduce that $f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$, which is also an open subset of X, so we conclude that f is continuous. □

Theorem 2.18. Let X, Y, and Z be topological spaces. If $f: X \to Y$ and $g: Y \to Z$ are continuous functions, then $g \circ f$ is continuous.

Proof. Let V be an open subset of Z. Since g is continuous, $g^{-1}(Z)$ is an open subset of Y. Following, since f is continuous, $f^{-1}(g^{-1}(Z))$ is an open subset of Z. Notice that $f^{-1}(g^{-1}(Z)) = (g \circ f)^{-1}(Z)$. Thus, $g \circ f$ is continuous. □

Definition 2.19 (Homeomorphism)**.** Let X and Y be topological spaces. A function $f: X \to Y$ is a *homeomorphism* if f is bijective, continuous, and f^{-1} is continuous i.e. a subset U is open in X if and only if $f(U)$ is open in Y. We say X and Y are *homeomorphic* if there exists a homeomorphism between X and Y.

Theorem 2.20. The property of being homeomorphic is an equivalence relation.

Proof. Reflexivity: Let X be a topological space. The identity map on X is a homeomorphism, so X is homeomorphic to itself.
Symmetry: Let X and Y be topological spaces and suppose X is homeomorphic to Y, so there exists a homeomorphism $f: X \to Y$. Since $f^{-1}: Y \to X$ is also a homeomorphism, Y is homeomorphic to X.
Transitivity: Let X, Y, and Z be topological spaces and suppose X is homeomorphic to Y and Y is homeomorphic to Z. Then, there exists homeomorphisms $f: X \to Y$ and $g: Y \to Z$. Since $g \circ f: X \to Z$ is also a homeomorphism, X is homeomorphic to Z. □

Definition 2.21 (Topological Property)**.** A property of a topological space is a *topological property* if it is preserved under homeomorphisms.

Some examples of topological properties include discreteness and metrizability. If X and Y are homeomorphic topological spaces, then X is discrete if and only if Y is discrete, and X is metrizable if and only if Y is metrizable.

The following theorem concerns a map from a topological space to a metric space.

Theorem 2.22. Let $(X, \mathcal{T})$ be a topological space and let (Y, d) be a metric space. A function $f: X \to Y$ is continuous at $x \in X$ if and only if for all $\varepsilon > 0$, there exists a neighborhood U of x such that $d(f(x), f(y)) < \varepsilon$ for any $y \in U$.

Proof. Suppose f is continuous at $x \in X$. Let $\varepsilon > 0$, and consider the open neighborhood $N_\varepsilon(f(x))$ in Y. Since f is continuous at x, there exists an open subset U of X such that $f(U) \subseteq N_\varepsilon(f(x))$.

Now, suppose the converse. Let $f(x) \in V$ for some open subset V of Y. Then, there exists $\varepsilon > 0$ such that $N_\varepsilon(f(x)) \subseteq V$, and by assumption, there exists an open subset U of X such that $f(U) \subseteq N_\varepsilon(f(x))$, so $f(U) \subseteq V$. □

3. BASE FOR A TOPOLOGY

In many cases, a topology is infinite and we cannot determine every set in a topology by hand. Thus, we are interested in a smaller collection of sets that satisfies a certain criterion that allows it to "generate" a topology. In particular, such a set is called the *base* of a topology.

Definition 3.23 (Base)**.** Let X be a topological space. A family $\mathcal{B}$ of open subsets of X is a *base* for the topology of X if every open subset of X is a union of sets in $\mathcal{B}$.

Theorem 3.24. A family $\mathcal{B}$ of open subsets of X is a base for the topology of X if and only if for each $x \in X$ and each neighborhood U of x, there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$.

Proof. Suppose $\mathcal{B} = \{V_\alpha\}_{\alpha \in A}$ is a base for the topology of X. Let $x \in X$, and let U be a neighborhood of X such that $x \in U$. Then, there exists an open subset S of X such that $x \in S \subseteq U$. Further, there exists a subset B of A such that $S = \bigcup_{\alpha \in B} V_\alpha$. Then, there exists $\alpha_0 \in B$ such that $x \in V_{\alpha_0} \subseteq S \subseteq U$.

Now, suppose that for each $x \in X$ and each neighborhood U of x, there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$. Then, it is clear that $U = \bigcup_{x \in U} V_x$ where V_x is an open subset of X such that $x \in V_x \subseteq U$ for each $x \in U$. Thus, we conclude that every open subset of X is the union of sets in $\mathcal{B}$, so $\mathcal{B}$ is a base for the topology of X. □

Theorem 3.25. A family $\mathcal{B}$ of open subsets of X is a base for the topology of X if and only if it satisfies the following two properties:

1. Every $x \in X$ lies in at least one set in $\mathcal{B}$.
2. If $x \in V \cap U$ for any sets $V, U \in \mathcal{B}$, there exists exists $W \in \mathcal{B}$ such that $x \in W \subseteq V \cap U$.

Proof. Suppose $\mathcal{B}$ is a base for the topology of X. Then, it immediately follows from Theorem 3.24 that (1) holds. Since $V \cap U$ is an open subset of X, it also immediately follows that (2) holds.

Now, suppose (1) and (2) hold. Let $\mathcal{T}$ be a topology for X. By (1), we deduce that X is the union of all sets of $\mathcal{B}$, so $X \in \mathcal{T}$, and it is clear that any union of sets in $\mathcal{T}$ is in $\mathcal{T}$. Then, it suffices to show that the intersection of any two subsets in $\mathcal{T}$ is in $\mathcal{T}$.

Let $U, V \in \mathcal{T}$. Since U and V are unions of sets in $\mathcal{B}$, it follows that there exists sets $F, G \in \mathcal{B}$ such that $x \in F \subseteq U$ and $x \in G \subseteq V$. Then, $x \in F \cap G$ and by (2), there exists $W_x \in \mathcal{B}$ such that $x \in W_x \subseteq F \cap G$, so $W_x \subseteq U \cap V$ as well. Then, $U \cap V = \bigcap_{x \in U \cap V} W_x$, so $U \cap V \in \mathcal{T}$. □

Recall the following definitions and theorems from section 1.4. The proofs for the theorems are identical to those in section 1.4.

Definition 3.26 (Second Countable)**.** Let X be a topological space. X is *second countable* if there is a countable family of open sets that forms a base for the topology for X.

Theorem 3.27 (Lindelöf's Theorem)**.** If X is a second countable topological space, then every open cover of X has a countable subcover.

Density and separability are two other definitions that are identical between metric spaces and topological spaces.

Theorem 3.28. If X is a second countable topological space, then X is separable.

Note that while the converse of this claim is true for metric spaces, it is *not* true for topological spaces in general.

4. SEPARATION AXIOMS

Recall that in the previous section, we were interested in topological spaces that have a minimal number of open sets. However, there are situations in topology where we are interested in maximal numbers of open sets in order to gain as much information about a structure as possible. *Separation properties* are properties that guarantee such an abundance of open sets.

Definition 4.29 (T_1-space)**.** A topological space X is a T_1*-space* if for any points $x, y \in X$, there exists an open set U such that $y \in U$ but $x \notin U$. Then, $X \setminus \{x\}$ is open since it can be expressed as the union of open sets, and thus $\{x\}$ is a closed subset of X.

Remark. It immediately follows that a topological space X is a T_1 space if and only if $\{x\}$ is closed for all $x \in X$.

Definition 4.30 (Hausdorff Space)**.** A space X is a *Hausdorff space* or a T_2*-space* if for any points $x, y \in X$, there exists disjoint open subsets U and V such that $x \in U$ and $y \in V$.

Remark. Clearly, every *Hausdorff space* is a T_1-space.

Definition 4.31 (Regular, Normal)**.** A space X is *regular* if for every closed subset E of X and for every point $x \in X \setminus E$, there exists *disjoint* open subsets U and V such that $U \subseteq E$ and $x \in V$.

A space X is *normal* if for every pair E and F of disjoint closed subsets of X, there exists *disjoint* open subsets U and V such that $U \subseteq E$ and $V \subseteq F$.

Definition 4.32 (T_3-space, T_4-space)**.** A space X is a T_3*-space* if it is a regular T_1-space. A space X is a T_4*-space* if it is a normal T_1-space.

Theorem 4.33. Every metric space is a T_4-space.

Proof. Let (X, d) be a metric space. Since $\{x\}$ is the intersection of closed sets $\overline{N_r(x)}$ for all $r > 0$, $\{x\}$ is closed so X is a T_1-space.

Let E and F be disjoint closed subsets of X. For each $x \in E$, there exists $r_x > 0$ such that $N_{r_x}(x) \cap F = \emptyset$, and for each $y \in F$, there exists $r_y > 0$ such that $N_{r_y}(y) \cap E = \emptyset$. Define $U := \bigcup_{x \in E} N_{r_x}(x)$ and $V := \bigcup_{y \in F} N_{r_y}(y)$.

We claim that U and V are disjoint. Arguing by contradiction, suppose there exists $z \in U \cap V$. Then, there exists $r_x > 0$ such that $d(x, z) < \frac{r_x}{2}$ for some $x \in E$ and there exists $r_y > 0$ such that $d(y, z) < \frac{r_y}{2}$ for some $y \in E$. Then,

$$d(x, z) \leq d(x, y) + d(y, z) < \frac{r_x}{2} + \frac{r_y}{2} \leq \max(r_x, r_y)$$

so either $x \in N_{r_y}(y)$ and $y \in N_{r_x}(x)$, a contradiction. Thus, we conclude that X is normal and thus, that X is a T_4-space. $\square$

Lemma 4.34. A topological space X is normal if and only if for each closed subset E of X and each open set W containing E, there exists an open set U containing E such that $\overline{U} \subseteq W$.

Proof. Suppose that X is normal. Let E be a closed subset of X and let W be a open subset of X containing E. Notice that E and $X \setminus W$ are disjoint closed subsets of X. Then, for any disjoint open subsets U and V of X such that U contains E and V contains $X \setminus W$, $\overline{U} \subseteq X \setminus V \subseteq W$.

Now, suppose the converse. Let E and F be disjoint closed subsets of X. Then, $W = X \setminus E$ is an open subset of X containing F. By assumption, there exists U, an open subset of X containing F such that $\overline{U} \subseteq W$. Then, $X \setminus \overline{U}$ and U are disjoint open subsets of X containing E and F respectively, so X is normal. □

The following two theorems are significant results that stem from normal topological spaces. In particular, the former, Urysohn's Lemma makes use of Dyadic rationals, rational numbers in the form $\frac{p}{2^n}$, where $p, n \in \mathbb{Z}$. While the proof is omitted here, the Dyadic rationals are dense in $\mathbb{R}$.

Theorem 4.35 (Urysohn's Lemma)**.** Let X be a *normal* topological space and let E and F be disjoint closed subsets of X. There exists a continuous function $f\colon X \to [0, 1]$ such that $f \equiv 0$ on E and $f \equiv 1$ on F.

Proof. Since X is a normal topological space and $X \setminus F$ is an open subset of X containing E, by Lemma 4.34, there exists an open set $U_{\frac{1}{2}}$ such that $E \subseteq U_{\frac{1}{2}} \subseteq \overline{U_{\frac{1}{2}}} \subseteq X \setminus F$. Similarly, there exist open sets $U_{\frac{1}{4}}$ and $U_{\frac{3}{4}}$ such that

$$E \subseteq U_{\frac{1}{4}} \subseteq \overline{U_{\frac{1}{4}}} \subseteq U_{\frac{1}{2}} \subseteq \overline{U_{\frac{1}{2}}} \subseteq U_{\frac{3}{4}} \subseteq \overline{U_{\frac{3}{4}}} \subseteq X \setminus F$$

Continuing, for each Dyadic rational $r \in (0, 1)$, we construct an open set U_r such that $\overline{U_r} \subseteq U_s$ for any Dyadic rational s such that $r < s$, $E \subseteq U_r$, and $U_r \subseteq X \setminus F$.

Following, we construct a function f such that $f(x) = 0$ if $x \in U_r$ for all $r \in (0, 1)$ and $f(x) = \sup\{r\colon x \notin U_r\}$. Thus, $f \equiv 0$ on E and $f \equiv 1$ on F.

It suffices to show that f is continuous. Let $x \in X$ such that $0 < f(x) < 1$ (it is easy to show continuity when $f \equiv 0$ or $f \equiv 1$). Let $\varepsilon > 0$, and let s and t be Dyadic rationals in $(0, 1)$ such that

$$f(x) - \varepsilon < s < f(x) < t < f(x) + \varepsilon.$$

Then, $x \notin U_r$ for any Dyadic rational $r \in (s, f(x))$ and also $x \notin \overline{U_r}$. Notice that $x \in U_t$, and thus that $W := U_t \setminus \overline{U_s}$ is an open neighborhood of x. Then, for any $y \in W$, $s < f(y) < t$, so $f(W) \subseteq N_\varepsilon(f(x))$, and thus, we conclude f is continuous. □

Theorem 4.36 (Tietze Extension Theorem)**.** Let X be a normal topological space, and let Y be a closed subset of X, and let f be a real valued function that is bounded and continuous on Y. Then, there exists a real valued function h that is bounded and continuous on X such that $h = f$ on Y.

5. COMPACTNESS

The definition of compactness for topological spaces is almost identical to that for metric spaces.

Definition 5.37 (Compactness)**.** Let X be a topological space. X is *compact* if every open cover of X has a finite subcover.

A subset S of a topological space X is a *compact subset* of S is compact in the relative topology that it inherits from X.

It may be helpful to note that in metric spaces and especially in real analysis, compactness is often defined with the assumption that a space is Hausdorff. In this section, we attempt to develop the concept of compactness without such an assumption.

Theorem 5.38. Any finite union of compact subsets of a topological space is compact.

Proof. Let $S_1, \dots, S_n$ be a finite family of compact subsets of a topological space X and let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of $\bigcup_{k=1}^n S_k$. For each $k \in \{1, \dots, n\}$, there exists a finite subset A_k of A such that $\bigcup_{\alpha \in A_k} G_\alpha$ covers S_k, and since a finite union of finite sets is finite, $\mathcal{A} := \bigcup_{k=1}^n A_k$ is finite. Then, since $\bigcup_{\alpha \in \mathcal{A}} G_\alpha$ covers $\bigcup_{k=1}^n S_k$, $\bigcup_{k=1}^n S_k$ is compact. □

Theorem 5.39. A closed subspace of a compact topological space is compact.

Proof. Let S be a closed subset of the compact subset X and let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of S. Notice that $\{G_\alpha : \alpha \in A\} \cup \{X \setminus S\}$ is an open cover of X, and since X is compact, there exists a finite subcover of $\{G_\alpha : \alpha \in A\} \cup \{X \setminus S\}$. Clearly, this finite subcover also covers S, so S is a compact subset of X. $\square$

Lemma 5.40. Let S be a compact subset of a Hausdorff space X. Then, for every $x \in X \setminus S$, there exists disjoint open neighborhoods U of x and V of S.

Proof. Since X is Hausdorff, for every $y \in S$, there exists disjoint open neighborhoods U_y of y and V_x of x. Notice that $\{U_y\}_{y \in Y}$ is an open cover of S, and since S is compact, there exists a finite subcover $\{U_{y_n}\}_{n=1}^N$. Then, $\bigcup_{n=1}^N U_{y_n}$ and V_x are disjoint open neighborhoods of S and x, respectively. $\square$